MATH 346/377 – Winter 2025

Prerequisites: A background in algebra (groups, rings, fields, and linear algebra) at the level of MATH 235

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Office Hours: TBD, see myCourses.

Required Course Materials: *Number Theory Revealed: A Masterclass* by Andrew Granville. Required for readings and problem sets. Ebook available for free through McGill library. For additional resources, see myCourses.

Course Topics: divisibilty, congruences, diophantine equations, arithmetic functions, quadratic reciprocity. If time permits we will also discuss: p-adic numbers, rational approximations, quadratic forms,..

Means of Assessment for Math 346: Your course grade will be the **maximum** of the following two assessment options:

Option 1: 20% assignments + 40% midterm + 40% final exam

Option 2: 20% assignments + 80% final exam

Means of Assessment for Math 377: Your course grade will be the **maximum** of the following two assessment options:

Option 1: 20% assignments + 20% project + 20% midterm + 40% final exam

Option 2: 20% assignments + 20% project + 60% final exam

Assessment Type	Description	Due Date	Considerations and Late Penalties
Midterm Exam	In-person; Crowdmark Long answer, proof-based questions	In person during class time (8:35 – 9:55) on **Wednesday, February 26.**	Students missing the midterm for any reason will automatically be assessed by assessment option 2
Final Exam	In-person; Crowdmark Long answer, proof-based questions	To be centrally scheduled during the final exam period (April 14 to April 30)	Missed final exams are handled by Service Point
Homework	6 graded assignments, to be submitted on Crowdmark A new assignment will be posted roughly every two weeks All assignments will feature long-answer, proof-based questions. **Your lowest of the 6 grades will be dropped.** Some problem sets will have some questions labeled "(377 only)." For MATH 346 students, only the questions without this label will be graded. For MATH 377 students, all questions will be graded.	Due dates are TBD, as class progresses.	Unexcused late assignments will not be accepted and will receive a grade of 0
Project(Math 377 only)	A paper on a topic related to (but going beyond) the course material. The final draft should be 10-15 pages.	Topic selection should be done by **Wednesday, February 19.** The final draft will be due on the last day of class, **Wednesday, April 9.**	Unexcused late projects will not be accepted and will receive a grade of 0

Rules and Regulations:

- The purpose of the Policy on Assessment of Student Learning (PASL) is to provide a set of common principles to guide the assessment of students' learning. Per PASL, assessment should be equitable and consistent, and promote effective learning experiences, a healthy learning environment, and academic integrity. Learn more on the For Students page of the PASL website. Also see Faculty of Science-specific rules on the implementation of PASL.
- Legally mandated academic accommodations are handled by Student Accessibility and Achievement. For more information see https://www.mcgill.ca/access-achieve/
- Academic considerations for assessments that are missed or late for valid reasons will be provided at the instructor's discretion. As per the new Quebec guidelines, medical notes are not required for absences of less than 5 days. *Note that repeated, similar requests for academic considerations in this course are unlikely to be granted.*
- In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French written work that is to be graded. This does not apply to courses in which acquiring proficiency in a language is one of the objectives. (Approved by Senate on 21 January 2009)

  Conformément à la Charte des droits de l'étudiant de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté, sauf dans le cas des cours dont l'un des objets est la maîtrise d'une langue. (Énoncé approuvé par le Sénat le 21 janvier 2009)
- McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures (Approved by Senate on 29 January 2003) (See McGill's guide to academic honesty for more information).
- In the event of extraordinary circumstances beyond the University's control, the content and/or assessment tasks in this course are subject to change and students will be advised of the change.